

Technical Document #347: Software Engineering - Comprehensive Overview

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Test-driven development (TDD) writes tests before writing code. It improves code quality and design through small, incremental changes.

Code review examines code changes before merging. It improves code quality, catches bugs, and shares knowledge across the team.

Microservices architecture structures applications as independently deployable services. Each service owns its data and business logic.

Refactoring improves code structure without changing behavior. It makes code easier to understand, maintain, and extend.

The Internet of Things (IoT) connects billions of devices worldwide. This connectivity generates massive amounts of data that can be analyzed for insights.

Federated learning trains models across decentralized data sources without sharing raw data. It enables privacy-preserving machine learning.

Computer vision applications are becoming ubiquitous, from facial recognition to autonomous driving. Deep learning has enabled breakthroughs in visual understanding.

Quantum computing promises to solve problems that are intractable for classical computers. It has potential applications in cryptography, drug discovery, and optimization.

Augmented reality overlays digital information on the physical world. It has applications in training, maintenance, and entertainment.

Data-driven decision making has become essential for modern businesses. Organizations that leverage data effectively gain a significant competitive advantage.

Key Takeaways:

- Test-driven development (TDD) writes tests before writing code.
- Refactoring improves code structure without changing behavior.
- Domain-driven design (DDD) models software around business domains.